

Non-Radial Fluid Oscillation

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1 Non-Radial Fluid Oscillation Modes of Compact Stars

In this section, we outline the equations governing the oscillation modes of the fluid comprising the neutron star matter. The most general metric for a spherically symmetric space-time is given by

$$\begin{aligned} ds^2 &= g_{\alpha\beta} dx^\alpha dx^\beta \\ &= e^{2\nu} dt^2 - e^{2\lambda} dr^2 - r^2(d\theta^2 + \sin^2\theta d\phi^2) \end{aligned} \quad (1)$$

where ν and λ are the metric functions. It is convenient to define the mass function, $m(r)$ in the favour of λ as

$$e^{2\lambda} = \left(1 - \frac{2m}{r}\right)^{-1}. \quad (2)$$

Starting from the line element Eq.(1) one can obtain the equations governing the structure of spherical compact objects, the Tolman-Oppenheimer-Volkoff (TOV) equations, as

$$\frac{dp}{dr} = -(\epsilon + p) \frac{d\nu}{dr} \quad (3)$$

$$\frac{dm}{dr} = 4\pi r^2 \epsilon \quad (4)$$

$$\frac{d\nu}{dr} = \frac{m + 4\pi r^3 p}{r(r - 2m)}. \quad (5)$$

In the above set of equations ϵ , p are the energy density and the pressure respectively. $m(r)$ is the mass of the compact star enclosed within a radius r . To solve these equations, one has to supplement these equations with an equation relating pressure and energy density *i.e.* an equation of state (EOS). Further, one has to set the boundary conditions at the center and surface as

$$m(0) = 0 \quad \text{and} \quad p(0) = p_c \quad (6)$$

$$p(R) = 0 \quad (7)$$

$$e^{2\nu(R)} = 1 - \frac{2M}{R}, \quad (8)$$

where the total mass of the compact object is given by $M = m(R)$ ¹, R being it's radius which is defined as the radial distance where the pressure vanishes while integrating out Eqs. (3, 4 and 5) from the center to the surface of the star. One can solve these equations along with a boundary conditions Eqs. (6, 7 and 8) for a set of central densities ϵ_c or corresponding pressure p_c to obtain the mass-radius, $(M - R)$ curve.

¹In this section, M denotes the mass of the compact stars to be distinguished from the constituent quark mass defined in Sec ??.

For the sake of completeness, we give below a succinct derivation of pulsating equations in the context of neutron star within a relativistic setting [?, ?]. The Einstein field equation that relates the curvature of space time to the energy momentum tensor is given as

$$R_{\alpha\beta} - \frac{1}{2}g_{\alpha\beta}R = 8\pi T_{\alpha\beta}, \quad (9)$$

with $T_{\alpha\beta}$ being the stress energy tensor, which for a perfect fluid is given by

$$T^{\mu\nu} = (p + \epsilon)u^\mu u^\nu - \epsilon g^{\mu\nu}, \quad (10)$$

with p and ϵ being the pressure and energy density respectively and u^μ is the four-velocity. Taking (covariant) divergence of the Einstein equation, Eq.(9), the left hand side of Eq.(9) vanishes using Bianchi identity leading to covariant conservation equation of the energy momentum tensor i.e. $T^{\mu\nu}_{;\mu} = 0$. With $T^{\mu\nu}$ given in Eq. (10), this reduces to

$$(p + \epsilon)u^\mu u_{\nu;\mu} = \partial_\nu p - u_\nu u^\mu \partial_\mu p \quad (11)$$

which is the relativistic Euler equation [?]. Next, to derive the equation of motion, we use the conservation of baryon number. This is similar to using continuity equation in non-relativistic case which follows from mass conservation. The baryon number conservation equation is given by

$$\frac{dn}{d\tau} = -n u^\mu_{;\mu}, \quad (12)$$

where, n is the baryon number density.

We shall derive the equations in spherical coordinates and the perturbations will be expanded in terms of vector spherical harmonics. The position (t, r, θ, ϕ) of a fluid element in space time as a function of proper time τ is given by the position four-vector $\xi(\tau)$ as

$$\xi(\tau) = \begin{pmatrix} \xi^t \\ \xi^r \\ \xi^\theta \\ \xi^\phi \end{pmatrix}. \quad (13)$$

Consider a fluid element located at ξ_0 as its equilibrium position is displaced to $\xi(\xi_0, \tau) = \xi_0 + \zeta(\xi_0, \tau)$. This results perturbation in pressure p , in energy density ϵ and in baryon number density n as

$$p = p_0 + \delta p \quad (14)$$

$$\epsilon = \epsilon_0 + \delta \epsilon \quad (15)$$

$$n = n_0 + \delta n \quad (16)$$

where, the subscript ‘0’ refers to the corresponding quantities in equilibrium. To derive the equations of motion for the perturbation, one has to linearize the Euler equation, Eq.(11) in the perturbation. For this we need the four velocities of the fluid elements $u^\mu = \frac{d\xi^\mu}{d\tau} = \frac{d\zeta^\mu}{d\tau}$. Further, we shall confine ourselves to performing the analysis for spherical harmonic component with the azimuthal index $m = 0$. For the displacement vector ζ^μ we take the ansatz

$$\begin{pmatrix} \zeta^t \\ \zeta^r \\ \zeta^\theta \\ \zeta^\phi \end{pmatrix} = \begin{pmatrix} t \\ \frac{e^{-\lambda} Q(r, t)}{r^2} P_l(\cos \theta) \\ -\frac{Z(r, t)}{r^2} \partial_\theta P_l(\cos \theta) \\ 0 \end{pmatrix}, \quad (17)$$

where $Q(r, t)$ and $Z(r, t)$ are the perturbing functions. We choose a harmonic time dependence for the perturbation *i.e.* $\propto e^{i\omega t}$ with frequency ω . Further, we do not consider here toroidal deformations. From the normalisation condition for the velocity $u_\mu u^\mu = 1$, and keeping upto linear terms in the perturbation, we have $u^t = d\zeta^t/d\tau = e^{-\nu}$. The other components of the four-velocity are given as

$$\begin{pmatrix} u^t \\ u^r \\ u^\theta \\ u^\phi \end{pmatrix} = \begin{pmatrix} e^{-\nu} \\ e^{-\nu} \dot{\zeta}^r \\ e^{-\nu} \dot{\zeta}^\theta \\ 0 \end{pmatrix}, \quad (18)$$

where, the dot on the perturbed coordinate denotes the derivative with respect to time ‘ t ’. Similarly, the contravariant velocity components are given using the metric given in Eq.(1) and Eq.(18) as

$$\begin{pmatrix} u_t \\ u_r \\ u_\theta \\ u_\phi \end{pmatrix} = \begin{pmatrix} e^\nu \\ -e^{2\lambda-\nu} \dot{\zeta}^r \\ -r^2 e^{-\nu} \dot{\zeta}^\theta \\ 0 \end{pmatrix}. \quad (19)$$

Now we simplify the Euler equation *i.e.* Eq.(11) by substituting the expressions for pressure, energy density and the fluid four-velocity and linearize in terms of the perturbing functions. The $\nu = t$ component of the Euler equation, Eq.(11), reduces to

$$(p_0 + \epsilon_0) \nu'(r) = -p'_0(r) \quad (20)$$

where the superscript ‘prime’ corresponds to derivative with respect to ‘ r ’. To obtain Eq.(20), we have used in the LHS of Eq.(11), with $\nu = t$, $u^\mu u_{t;\mu} = \nu' \dot{\zeta}^r$ and in RHS we have used the fact that p_0 is isotropic so that $\dot{p}_0 - u_t u^\mu \partial_\mu p \sim -\dot{\zeta}^r p'_0(r)$. Let us recognise that the Eq.(20) is essentially a part of the TOV equations (Eq.(3)) relating pressure gradient and the metric function gradient. Next, the $\nu = r$ component of the Euler equation, Eq.(11), reduces to

$$\omega^2(\epsilon_0 + p_0) e^{2(\lambda-\nu)} \dot{\zeta}^r - (\delta\epsilon + \delta p) \nu'(r) - \frac{d}{dr}(\delta p) = 0 \quad (21)$$

Similarly, the $\nu = \theta$ component of the Euler equation, Eq.(11), by using $u^\mu u_{\theta;\mu} = u^t \partial_t u_\theta = -e^{-2\nu} r^2 \ddot{\zeta}^\theta$, is given as

$$\omega^2(\epsilon_0 + p_0) e^{-2\nu} r^2 \ddot{\zeta}^\theta - \partial_\theta \delta p = 0. \quad (22)$$

Having written down the Euler equation to linear order in the perturbation, let us next consider the baryon number conservation equation *i.e.* Eq.(12). With the velocity components given in Eqs.(18, 19) and Eq.(17) for the perturbation, the number conservation equation, Eq.(12) can be written in terms of the radial and azimuthal perturbing functions $Q(r)$ and $Z(r)$ as

$$\frac{dn}{d\tau} = -\frac{n}{r^2} \left[e^{-(\lambda+\nu)} \frac{\partial^2 Q(r, t)}{\partial r \partial t} + e^{-\nu} l(l+1) \dot{Z} \right] P_l(\cos \theta) \quad (23)$$

We might note here that, since the proper time derivative is taken along the world line of the fluid parcel, we can write $\frac{dn}{d\tau} = \frac{d\Delta n}{d\tau}$, where, Δn is the Lagrangian perturbation. Further, using the relation $\partial/\partial t = e^{-\nu} \partial/\partial \tau$, we can integrate Eq.(23) over $d\tau$ to obtain the Lagrangian perturbation in number density Δn in terms of the perturbing functions Q and Z as

$$\frac{\Delta n}{n_0} = -\frac{1}{r^2} [e^{-\lambda} Q' + l(l+1)Z] P_l(\cos \theta) \quad (24)$$

To write down the equations in terms of the perturbing functions $Q(r)$ and $Z(r)$, we need to express the energy density perturbation $\delta\epsilon$ and pressure perturbation δp occurring in Eqs.(20, 21) in terms of the functions $Q(r)$ and $Z(r)$. The strategy is to use the Euler equation Eq.(11) to write $\delta\epsilon$ in terms of δn and use definition of bulk modulus ($\kappa = n \frac{\Delta p}{\Delta n}$) to write δp in terms of δn . One can then use the baryon number conservation equation Eq.(23) to write $\delta\epsilon$ and δp in terms of the perturbing functions.

Thus, using the Euler equation Eq.(11) to eliminate $u_{;\mu}^\mu$ in the baryon number conservation Eq.(12), we have

$$\frac{dn}{d\tau} = \frac{n}{p + \epsilon} \frac{\partial \epsilon}{\partial \tau}, \quad (25)$$

which leads to

$$\Delta \epsilon \simeq \frac{\epsilon_0 + p_0}{n_0} \Delta n. \quad (26)$$

Further, using the relation between the Lagrangian perturbation and the Eulerian perturbation *i.e.* $\Delta \epsilon = \delta \epsilon + \zeta^r \frac{d\epsilon_0}{dr}$ and using Eq.(24), we have

$$\delta \epsilon = - \left[\frac{\epsilon_0 + p_0}{r^2} \{e^{-\lambda} Q' + l(l+1)Z\} + \frac{e^{-\lambda}}{r^2} Q \frac{d\epsilon_0}{dr} \right] P_l(\cos \theta) \quad (27)$$

Next, let us find out the relation between δp and Δn . The Eulerian variation δp and the Lagrangian variation Δp are related as

$$\delta p = \Delta p - \zeta^r \frac{dp_0}{dr} \quad (28)$$

Thus using Eq.(17) and Eq.(24), we have

$$\delta p = - \left[\frac{\kappa}{r^2} (e^{-\lambda} Q' + l(l+1)Z) + \frac{e^{-\lambda}}{r^2} \frac{dp_0}{dr} Q \right] P_l(\cos \theta) \quad (29)$$

Further Δp is related to Δn , through bulk modulus κ *i.e.*

$$\kappa = n \frac{\Delta p}{\Delta n}.$$

In the relativistic Cowling approximation, the metric perturbations are neglected. This will mean the energy and pressure perturbations should also vanish. In the relativistic Cowling approximation, the energy density perturbation $\delta\epsilon$ is set to zero but pressure perturbation is not set to zero. As shown in Ref.[?], such an approximation leads to qualitatively correct result which we shall also follow. Setting $\delta\epsilon = 0$ in Eq.(21), and using Eq.(29), we have

$$\begin{aligned} \nu' \delta p + \frac{d\delta p}{dr} = & -\nu' \kappa X - \frac{d(\kappa X)}{dr} - \nu' (p_0 + \epsilon_0) l(l+1) \frac{Z}{r^2} \\ & + (p_0 + \epsilon_0) Q \frac{d}{dr} \left(\frac{e^{-\lambda} \nu'}{r^2} \right), \end{aligned} \quad (30)$$

where we have defined for the sake of brevity $X = (e^{-\lambda} Q' + l(l+1)Z)/r^2$. Using this, the radial Euler equation, Eq.(21) becomes

$$\begin{aligned} \omega^2 (\epsilon_0 + p_0) e^{\lambda-2\nu} \frac{Q}{r^2} + \frac{d[\kappa X]}{dr} \\ + \nu' \kappa X + \nu' (\epsilon_0 + p_0) l(l+1) \frac{Z}{r^2} \\ - (\epsilon_0 + p_0) \frac{d}{dr} \left(\frac{e^{-\lambda} \nu'}{r^2} \right) = 0 \end{aligned} \quad (31)$$

Similarly, the azimuthal component of the Euler equation Eq.(22) becomes

$$\omega^2(p_0 + \epsilon_0)e^{-2\nu}Z - \kappa X - p'_0 \frac{e^{-\lambda}Q}{r^2} = 0 \quad (32)$$

It can be shown that the Eq.(31) through a rearrangement of terms is identical to that obtained earlier by McDermott *et al* [?] with an appropriate change of factor 2 in the metric functions $\nu(r)$ and $\lambda(r)$. Few more comments here may be in order. In literature, sometimes the adiabatic index γ is used instead of κ and is defined as [?]

$$\gamma = \left(\frac{\partial \ln p_0}{\partial \ln n_0} \right)_s = \frac{n_0 \Delta p}{p_0 \Delta n} \quad (33)$$

so that $\kappa = \gamma p_0$. Further, the same can be related to adiabatic speed of sound as follows. By using the definition of Jacobian and standard thermodynamic relation

$$\left(\frac{\partial \ln p_0}{\partial \ln n_0} \right)_s = \frac{n_0^2}{p_0 \chi_{\mu\mu}} \quad (34)$$

in the zero temperature limit. The adiabatic speed of sound at zero temperature is defined as [?]

$$c_s^2 = \left(\frac{\partial p_0}{\partial \epsilon_0} \right)_s = \frac{n}{\mu \chi_{\mu\mu}}$$

so that

$$\gamma = \frac{p_0 + \epsilon_0}{p_0} c_s^2 \quad (35)$$

Let us note that Eq.(31) is a second order differential equation for the perturbing function $Q(r)$. We now use Eq.(32) to write down two coupled first order equation for the perturbing functions. Using Eq.(32) and Eq.(35), we have the equation for perturbation as

$$Q' - \frac{1}{c_s^2} [\omega^2 r^2 e^{\lambda-2\nu} Z + \nu' Q] + l(l+1)e^\lambda Z = 0 \quad (36)$$

Next one can calculate the combination $d[Eq.(32)]/dr + [Eq.(31)]$ and substitute Eq.(32) again which leads to the first order differential equation for Z' as

$$Z' - 2\nu' Z + e^\lambda \frac{Q}{r^2} - \nu' \left(\frac{1}{c_e^2} - \frac{1}{c_s^2} \right) \left(Z + \nu' e^{-\lambda+2\nu} \frac{Q}{\omega^2 r^2} \right) = 0 \quad (37)$$

In the above equation $c_e^2 = \frac{dp_0}{d\epsilon_0} = \frac{p'_0}{\epsilon'_0}$ is the equilibrium speed of sound. It may be noted that Eq.(40) can be rewritten as

$$\omega^2 e^\lambda \frac{Q}{r^2} + \omega^2 Z' + A_- e^\lambda \omega^2 Z - A_+ e^{2\nu} \frac{p'_0}{p_0 + \rho_0} \frac{q}{r^2} = 0 \quad (38)$$

where, $A_+ = e^{-\lambda}(\epsilon'_0/(p_0 + \epsilon_0) + \nu'/c_s^2)$ and $A_- = A_+ - 2\nu'e^{-\lambda}$. It is reassuring to see that the Eq. (36) and Eq. (38) are identical to the corresponding equations Eq.(3b) and Eq.(4a) given in Ref. [?]. The gravity mode (g mode) oscillation frequencies are closely related to the Brunt-Väisälä frequency, ω_{BV} [?]. The relativistic generalisation of ω_{BV} is given by

$$\omega_{BV}^2 = \nu'^2 e^{2\nu} \left(1 - \frac{2m}{r} \right) \left(\frac{1}{c_e^2} - \frac{1}{c_s^2} \right) \quad (39)$$

This also reduces to the expression for the ω_{BV} in Newtonian limit [?].

The equation for the perturbation function $Z(r)$ can be rewritten in terms of the Brunt-Väisälä frequencies as

$$Z' - 2\nu'Z + e^\lambda \frac{Q}{r^2} - \frac{\omega_{BV}^2 e^{-2\nu}}{\nu' \left(1 - \frac{2m}{r}\right)} \left(Z + \nu' e^{-\lambda+2\nu} \frac{Q}{\omega^2 r^2} \right) = 0 \quad (40)$$

The two coupled first order differential equations for the perturbing functions $Q(r, t)$ and $Z(r, t)$, Eqs.(36),(40), are to be solved with appropriate boundary conditions at the center and the surface. Near the center of the compact stars the behavior of the functions $Q(r)$ and $Z(r)$ are given by [?]

$$Q(r) = Cr^{l+1} \quad \text{and} \quad Z(r) = -Cr^l/l \quad (41)$$

where C is an arbitrary constant and l is the order of the oscillation. The other boundary condition is the vanishing of the Lagrangian perturbation pressure, *i.e.* $\Delta p = 0$ at the stellar surface. Using equations Eqs.(28, 29 and 36), we have the Lagrangian perturbation pressure Δp given as

$$\Delta p = -\frac{(p_0 + \epsilon_0)}{r^2} [\omega^2 r^2 e^{\lambda-2\nu} Z + \nu' Q] e^{-\lambda} \quad (42)$$

Thus the vanishing of Δp at the surface of the star ($r = R$) leads to the boundary condition [?]

$$\omega^2 r^2 e^{\lambda-2\nu} Z + \nu' Q \Big|_{r=R} = 0 \quad (43)$$

Further, in case one considers stellar models with a discontinuity in the energy density, one has to supplement additional condition at the surface of discontinuity demanding Δp to be continuous *i.e.* $\Delta p(r = r_{c-}) = \Delta p(r = r_{c+})$. Where, r_c is the radial distance of the surface of energy density discontinuity from the center. This leads to [?, ?]

$$Q_+ = Q_-, \quad (44)$$

$$Z_+ = \frac{e^{2\nu}}{\omega^2 r_c} \left\{ \frac{\epsilon_{0-} + p_0}{\epsilon_{0+} + p_0} \left(\omega^2 r_c^2 e^{-2\nu} Z_- + e^{-\lambda} \nu' Q_- \right) - e^{-\lambda} \nu' Q_+ \right\}, \quad (45)$$

where the $-(+)$ subscript corresponds to the quantities before(after) the surface of discontinuity. In case of a Maxwell construct for phase transition, there is a discontinuity in energy density while in Gibbs construct of phase transition the energy density is continuous at the phase boundary as considered here.

With these boundary conditions the problem becomes an eigen-value problem for ' ω '. To calculate the eigen frequencies ω , we proceed as follows. For a given central density ϵ_c , we first solve the TOV equations Eqs.(3 - 5) to get the profile of the unperturbed metric functions $\lambda(r)$, $\nu(r)$ and also the mass M and the radius R of the spherical star. For a given ω , we solve the pulsating equations Eqs.(36 and 40) to determine the fluid perturbing functions $Q(r)$ and $Z(r)$ as a function of r . To solve these equations, we take the initial values for Q and Z consistent with Eq.(41). Specifically we took C of the order 1. The solutions of Q and Z are independent of this choice. We then calculate LHS of Eq.(43). The value of ω is then varied such that the boundary condition, Eq.(43), is satisfied. This gives the frequency, ω as function of mass and radius. It may be noted that there can be multiple solutions of ω satisfying the pulsating equations and the boundary conditions corresponding to different initial trail values for ω . These different solutions for ω correspond to frequencies of different modes of oscillations of the compact star.